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LIBERIA

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FOURTH REVIEW UNDER THE EXTENDED CREDIT FACILITY ARRANGEMENT AND REQUESTS FOR WAIVERS OF NONOBSERVANCE OF PERFORMANCE CRITERIA, MODIFICATION OF PERFORMANCE CRITERIA, AND REPHASING AND EXTENSION OF THE ARRANGEMENT—DEBT SUSTAINABILITY ANALYSIS[1]

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Liberia's risk of external debt distress has increased from low to moderate. The combined impacts of the Ebola crisis and the sharp decline in commodity prices have significantly affected the growth and exports outlook, contributing to deteriorated debt sustainability. Under the baseline scenario, all debt indicators are below their policy-dependent thresholds. However, due to lower nominal GDP and exports and higher new borrowings compared with the previous DSA update, stress tests indicate that the present value of public external debt-to-GDP and debt-to-exports indicators would breach the threshold over the medium term under the most extreme shock scenario. The debt outlook under alternative scenarios also highlights continued vulnerabilities to negative shocks to borrowing terms, the exchange rate, exports and growth. Hence, it would remain important for the authorities to continue to prioritize grants and highly-concessional financing while developing new medium term borrowing plans consistent with preserving debt sustainability.

[1] The LIC-DSA incorporates the following general assumptions: (i) the discount rate is fixed at 5 percent; (ii) the exchange rates are based on WEO assumptions; and (iii) the risk of debt distress based on country-specific policy-dependent thresholds, based on the country's CPIA index, which for Liberia is 3.09.

INTRODUCTION

1. **This debt sustainability analysis (DSA) is a full joint Bank-Fund LIC-DSA; the last LIC-DSA (an update) was considered by the Executive Board in July 2014 as part of Liberia's third review under the Extended Credit Facility Arrangement (ECF)**.[2] The heavy toll of the Ebola outbreak, compounded by the negative impact of the sharp decline in iron ore prices, have led to significantly lower growth and higher external borrowings in 2014–15, as well as a worsened medium term outlook. Following the large impact of these twin shocks, key external debt indicators, especially the debt-to-exports and debt-to-GDP ratio, have deteriorated over the medium term. The current DSA results suggest a moderate risk rating of debt distress for Liberia, compared with the low risk rating reported in the last DSA update in July 2014. Liberia's capacity to monitor debt continues to be assessed as being adequate.

BACKGROUND

2. **Liberia was hit by two large negative shocks in 2014, with significant implications on the economic outlook**. The Ebola outbreak since mid-2014 has severely impacted the economy, with a particularly heavy toll on agriculture and services, and the impact is expected to linger over a longer period. Notwithstanding that, the sharp decline in the price of key export commodities, most notably the 60 percent drop of iron ore prices since early 2014, has affected exports and caused delays in investment in the sector. As a result, growth in 2014–15 was revised down from 8–9 percent at the time of the 3rd ECF review to the current 0–1 percent. For 2016–18, average annual growth was revised downward from 8 ¾percent pre-crisis to about 5 percent.

3. **The pace of external borrowing has accelerated in recent years in line with the authorities' poverty reduction strategy–the Agenda for Transformation, alongside additional Ebola-related loan support in 2014–15**. After Liberia reached the HIPC initiative completion point in June 2010, external debt sustainability improved, providing fiscal space for new government borrowing to finance key public investment projects. After a slow start, the pace of external borrowing has accelerated since 2012 and total newly contracted external borrowings amounted to US$843.5 million (42 percent of GDP) as of end-July 2015, mainly on account of increased external financing for infrastructure and energy projects, with about 65 percent of the total funding provided by traditional multilateral creditors on highly concessional terms. As a result, the annual average PV of total new borrowings amounted to 8.3 percent of GDP over the program period to end-June 2015, higher than the 5.3 percent target agreed at the time of the 3rd ECF review, of which the Ebola-related borrowings and lower GDP contribute about one percentage point.

[2] The DSA was prepared jointly by the staff of the IMF and Bank, in collaboration with the authorities of Liberia. The last joint DSA update prepared for the 3rd ECF review can be found in IMF Country Report No. 14/197, July 2014.

4. **In light of the worsened economic outlook, debt sustainability has deteriorated**. Although public external debt remained relatively low at about 23 percent of GDP at June-2015, it is expected to increase steadily over the medium term with the disbursement of the recently-ratified new borrowings.

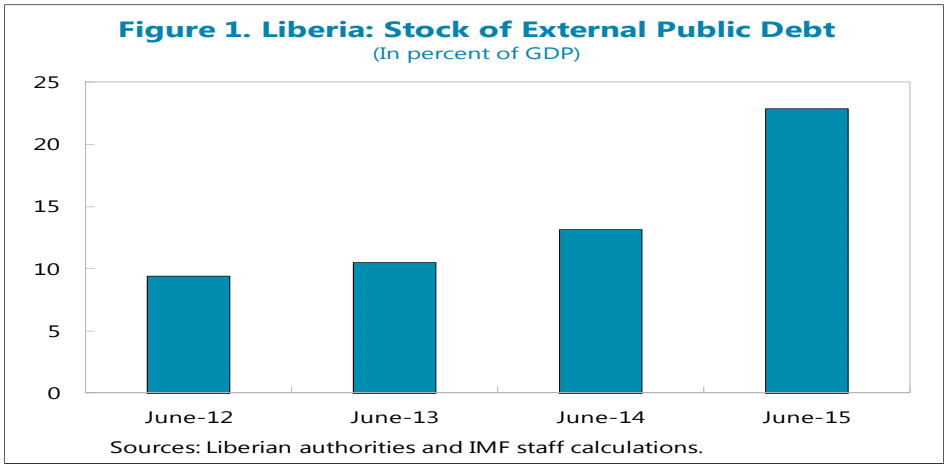


Table 1. Liberia: Structure of External Public Debt as of June-2015[1/]

	USD millions	Percent of Total	Percent of GDP
Multilateral	**425**	**93**	**21**
IMF	154	34	8
World Bank	159	35	8
AfDB	33	7	2
EIB	33	7	2
Other Multilateral	46	10	2
Bilateral:Non-Paris Club	**33**	**7**	**2**
China	6	1	0
Kuwait	9	2	0
Saudi Arabia	19	4	1
Total	**458**	**100**	**23**

Sources: Liberian authorities and IMF staff calculations.
[1] Debt stock on disbursement base, excluding debt to French C2D as the debt repayment will be returned to Liberia as grants for development projects and debt to Taiwan Province of China (85.4 mil) that is not servicing due to the lack of diplomatic relationship.

UNDERLYING ASSUMPTIONS

5. **The baseline macroeconomic assumptions underlying this DSA are summarized in Box 1**. Specifically, in staff's baseline projection, economic performance during 2014–15 reflects the twin shocks of the Ebola outbreak and the sharp decline in the prices of key export commodities. Medium and long-term macroeconomic assumptions rely heavily on the sustained containment of the Ebola outbreak, the expansion of large mining projects starting from 2020 (delayed, and on a lower scale than anticipated at the time of the 3rd ECF review), and the continuation of sound macroeconomic policies.

6. **The key changes in the baseline macroeconomic assumptions relative to the previous DSA update are as follows**:

- Real GDP growth is projected to be much lower during 2014–15 and over the medium term, reflecting the negative impact of the twin shocks. The GDP deflator is also revised downward significantly to reflect recent price developments in the iron ore sector.[3]

- The overall fiscal deficit in FY 2015–16 is projected to be higher than envisaged in the previous DSA, driven by higher current expenditure, and capital spending related to the Mount Coffee hydroelectric project and the implementation of the Economic Stabilization and Recovery Plan (ESRP). In the medium to long term, public expenditure is expected to gradually slow down, with fiscal balance as percent of GDP in line with previous projections.

- External new debt disbursements during FY2015–18 are higher reflecting the disbursement of higher-than-anticipated new external government borrowing already ratified or in the process of being ratified in FY2015. Over the longer run, external borrowing as percent of GDP is assumed to be in line with the last DSA update.

- The DSA incorporates the ECF augmentation of SDR 32.3 million, the debt relief under the CCR Trust of SDR 25.84 million and the RCF disbursement of SDR 32.3 million. The current DSA also takes into account the authorities' latest debt stock data as of June-2015 while excluding Taiwan Province of China and French CD2 loans that are expected to be fully restructured.

- The current account deficit is significantly higher in 2015 compared with the previous DSA update, reflecting the impact of the commodity price decline. Both exports and imports of goods are significantly lower in the near to medium term compared with levels envisaged at the time of the previous DSA update. While services imports and current transfers are much larger in FY2014/15 reflecting Ebola-related international support and are projected to return to the pre-crisis level over the medium term.

Table 2. Major Changes in DSA Assumptions
(Average over the 20-year projection period)

	3rd Review	4th Review
Real GDP growth (percent)	6.2	5.6
GDP deflator in USD (in percent)	3.3	2.4
Current account deficit (percent of GDP)	-22.4	-31.8
Primary deficit (percent of GDP)	-2.4	-2.7

[3] Iron ore deflator decreased more than 55 percent from 2014 to 2015, contributing roughly -4 percent to changes in GDP deflator, while changes in non-iron GDP deflator is about 3.7 percent.

Box 1. Key Baseline Macroeconomic Assumptions, 2015–35[1/]

Real GDP growth. Output is projected to decline by 2.7 percent in FY2015, reflecting the heavy toll of the Ebola outbreak on most economic sectors and the impact of the sharp decline in the prices of iron ore and rubber exports. With economic activity gradually returning to normal following the eradication of Ebola, real GDP growth is projected to increase to 4–5 percent in FY2016 and FY2017, buoyed by recovery in agriculture and services, and the coming on stream of a new gold mine. Growth is expected to average 5.4 percent in FY2016–20, lower than the projection at the time of the third ECF review (pre-Ebola and pre-commodity price shock). Long term growth over FY2021–35 is projected to average around 6.1 percent as mining production reaches full capacity and non-mining activity stabilizes. There are significant downside risks to the growth projection, including a prolonged period of low commodity prices, a flare-up of the Ebola disease as well social unrest as UNMIL is set to fully withdraw by end-2016 before the 2017 Presidential election.

Inflation. Inflation is projected to remain relatively high at around 8-8 ½ percent in FY2015–16 as the rebound in activity and imports offsets the impact of low international oil prices. In the long run, inflation is projected to gradually decline and stabilize at around 5 percent.

External account. The significant fall in iron ore and rubber prices will adversely affect exports growth in the near and medium term, although this could be partly offset by the coming on stream of a new gold mine by end-2015. Hence, exports are projected to fall by 16 percent in FY2015 and further decline by 12 percent in FY2016. Beyond, annual export growth is projected to gradually recover over the medium term and stabilize at around 5 percent toward 2035. Imports are projected to increase by 5 percent in FY2015 as increased demand combined with aid-related inflows is expected to outweigh the fall in fuel imports. Imports are projected to contract in FY2016–18 owing mainly to the expected drawdown of UNMIL before recovering to an average 8 percent growth during FY2019–24 largely driven by imports of capital goods related to the expansion in the iron ore sector. Beyond 2024, these effects would be phased out and the annual growth in goods and services imports is expected to average 5 percent. With the end of the Ebola epidemic, donor transfers are expected to fall sharply after 2015 from 60 percent of GDP to about 30 percent of GDP in FY2020 and to gradually decline to 13 percent of GDP by the end of the projection period. As a result, the current account deficit is projected to widen further in FY2015 to 40 percent of GDP but will gradually decline to reach 25 percent of GDP by FY2035, in line with the historical average level prior to the recent Ebola and commodity price shocks.

Tax revenues. The tax revenue to GDP ratio is projected to improve from 18.6 percent in FY2015 to around 21 percent in FY2021 and remain broadly stable at about 21 percent going forward.

External borrowing. Loans ratified since 2012 totaled US$843.5 million as of end-July 2015, most of which are on highly concessional terms for priority infrastructure projects, with another US$180 million projects in the pipeline for FY2016. The annual average NPV of signed external borrowing and pipeline loans is projected to be 7.1 percent of GDP over the program period up to June-2016, higher than the 5.3 percent target agreed in the IMF 3rd ECF review in July 2014. The expected disbursement of recently ratified loan agreements over the medium term, compounded by ambitious investment plans under the Economic Stabilization and Recovery Plan (ESRP), would raise external debt from about 14 percent of GDP in FY2014 to a projected 40 percent of GDP in FY2020. Beyond this, disbursement of external borrowing is expected to gradually decline from 4 percent of GDP in FY2021 to 2.5 percent of GDP toward the end of the projection period, in line with historical average. Most of the new external borrowings are assumed to be on highly concessional terms before 2020, except the US$14.2 million loans already signed on non-concessional terms. Starting from 2021, a small portion of borrowings are assumed on non-concessional terms from Chinese banks.

[1/] All data refer to fiscal year which runs from July to June.

EXTERNAL DEBT SUSTAINABILITY ANALYSIS

7. **The external DSA results indicate an increased risk of debt distress over the medium term** (Figure 2 and Table 3). In the medium term, the PV of debt-to-GDP ratio is projected to rise steadily from 14 percent in FY2015 to 23 percent by FY2020 in the baseline scenario, and gradually decline thereafter. With the deteriorated exports outlook, the PV of debt-to-export ratio is projected to increase significantly from 40 percent in FY 2015 to a peak of 98 percent in FY 2020, very close to the 100 percent threshold. Due to the highly concessional nature of new debt, debt-to revenues and debt service indicators remain well below the thresholds. These thresholds are based on the country policies and institutions assessment (CPIA) compiled annually by the World Bank.

8. **Compared with the previous DSA, the baseline debt indicators have worsened over the medium term**. In the baseline scenario, all debt indicators remain below the policy-dependent thresholds. The lower level of nominal GDP, lower exports and higher external borrowing result in a worsening of key debt indicators over the medium term. This highlights the relatively limited scope for the government to take on more external debt during the next five years, before the projected rise in mining sector exports and fiscal revenues, and the continued need for seeking concessional external resources. In light of the worsened borrowing capacity, the authorities indicated during the 4th review mission in October 2015 that they are streamlining their medium term borrowing plan and reviewing the current debt strategy. They noted that they intend to develop a new medium-term debt strategy with technical support from the IMF and the World Bank before the end of FY2016.

9. **The sensitivity analysis indicates deteriorated debt-to-GDP and debt-to-exports indicators, breaching the policy-dependent threshold under the extreme shock scenario**.

- **PV of external debt-to-GDP ratio**. Under the extreme shock scenarios of one-time large nominal depreciation, the PV of external debt-to-GDP breaches the 30 percent threshold throughout FY2018–23. The historical scenario shows that if key macroeconomic variables return to their historical average between FY2008 to FY2014, PV of debt increases from around 14 percent of GDP to 24 percent of GDP towards the end of the projection period.
- **PV of external debt and debt service-to-exports ratio**. Under the extreme shock scenarios of less favorable borrowing terms, the PV of external debt-to-exports ratio breaches the 100 percent threshold from FY2018 onwards and reaches a peak of 129 percent in FY2020, reflecting worsened exports outlook related to weak commodity prices. The debt service ratio remains well below the threshold of 15 percent in all scenarios.
- **PV of external debt and debt service-to-revenue ratio**. The PV of external debt-to-revenue ratio shows sensitivity to less favorable borrowing terms and one-time large depreciation. Both the debt and debt service-to-revenue ratios are well below the policy thresholds in all scenarios throughout the projection period.

10. **The probability approach further confirms the moderate risk rating of external debt distress**. According to the LIC Debt Sustainability Framework guideline, the evolution of debt-exports ratio in Figure 2 suggests a "moderate-to-high borderline case" where the debt-to-exports ratio fall within a

10-percent band around the indicative threshold in the baseline scenario, triggering the use of the probability approach as an alternative methodology for assessing the risk of external debt distress. As shown in Figure 4, the projected probability of debt distress for the debt-to-exports ratio remains within the threshold in stress scenarios while the probability of debt-to-GDP ratio breaches the threshold by more than 10 percent and lasts for about 6 years during 2018–23. This combined with the results shown in Figure 2 confirm a moderate risk rating of debt distress.

PUBLIC DEBT SUSTAINABILITY ANALYSIS

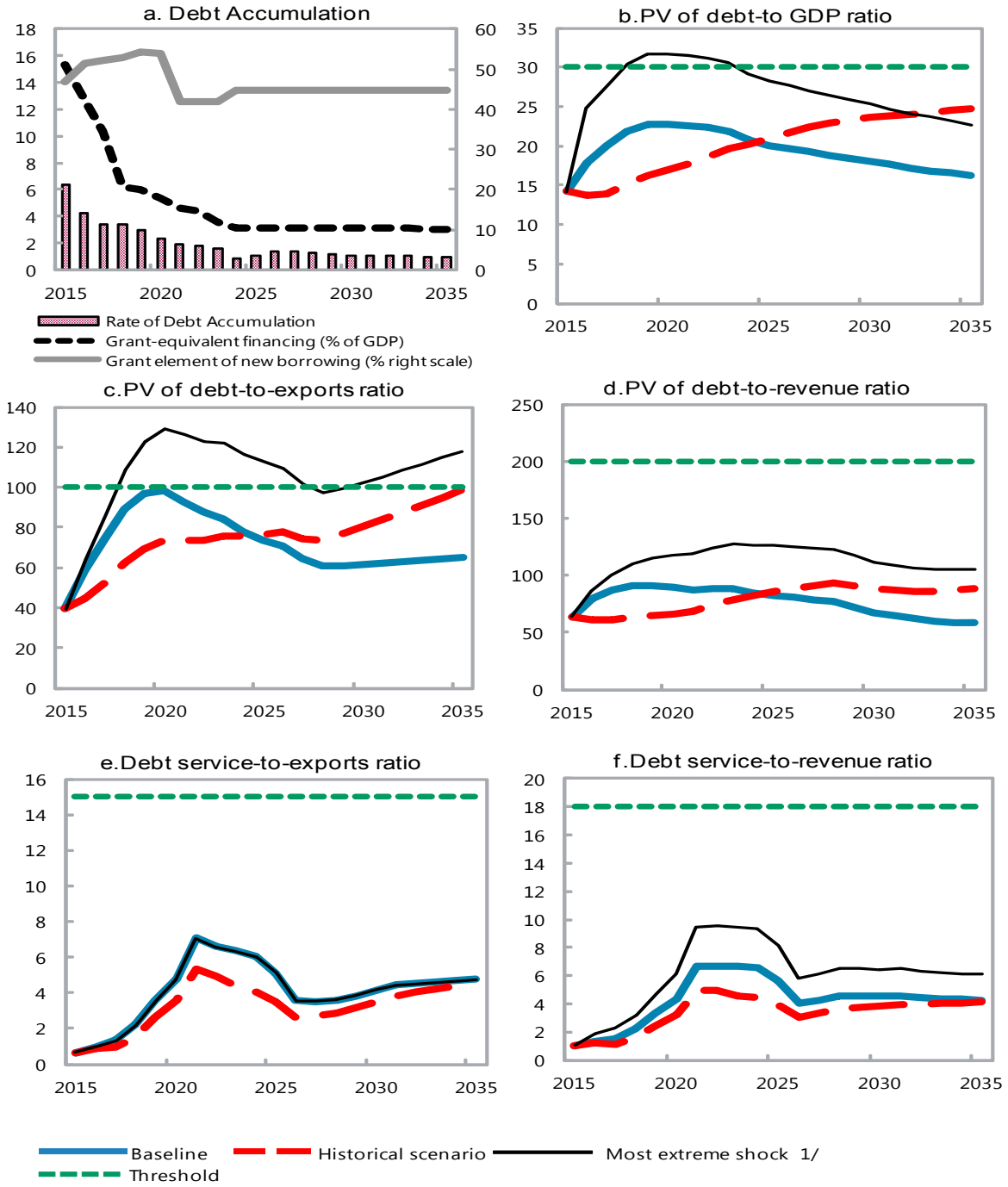
11. **The inclusion of domestic debt in the DSA worsens the debt burden indicators compared to the previous DSA** (Figure 3, Tables 5). Under the baseline and shock scenarios, all debt indicators have deteriorated in the near and medium term, though they remain below their respective thresholds. Under the baseline, the PV of public debt-to-GDP rises steadily from 15 percent in FY2015 to around 24 percent in FY2020 and is expected to fall afterwards. The PV of debt-to-revenue ratio rises to a peak of 84 percent in FY2018 and then follows a gradual downward path towards the end of the projection period. The PV of debt service-to-revenue ratio remains below the threshold afterwards.

12. **Stress tests highlight the risks associated with adverse macroeconomic shocks** (Figure 3, Table 6). Public debt indicators are particularly sensitive to lower growth, a large real depreciation and a significant increase in other debt-creating flows. Under the alternative scenario of a 10 percent increase in other debt-creating flows in FY2016, the PV of debt-to-GDP ratio will increase from 15 percent in FY2015 to the peak at about 29 percent by FY2018. The PV of the public debt-to-revenue ratio also deteriorates under this scenario, reaching over 100 percent around FY2018/19. However, the debt service-to-revenue ratio will remain below 15 percent under the alternative scenario.

CONCLUSION

13. **The projected increase in Liberia's debt over the medium term indicates a moderate risk of debt distress**. This has been revised up from the low risk rating under the previous DSA update in July 2014 mainly owing to the unexpected two large shocks experienced since 2014. The underlying macroeconomic assumptions and DSA results were discussed and agreed with the authorities during the mission. In the baseline scenario, all debt indicators remain below their respective thresholds. However, stress tests indicate that Liberia's external debt outlook has deteriorated over the medium term and continues to be vulnerable to macroeconomic shocks, in particular to changes in borrowing terms, growth, exports, and exchange rate depreciation. The inclusion of domestic debt leads to a deterioration in debt indicators. The increased risk of debt distress highlights the need for sound macroeconomic policies and prudent debt management, including relying primarily on concessional financing. The authorities' plan to review the current debt management strategy would be important to develop new medium to long term borrowing plans consistent with preserving debt sustainability.

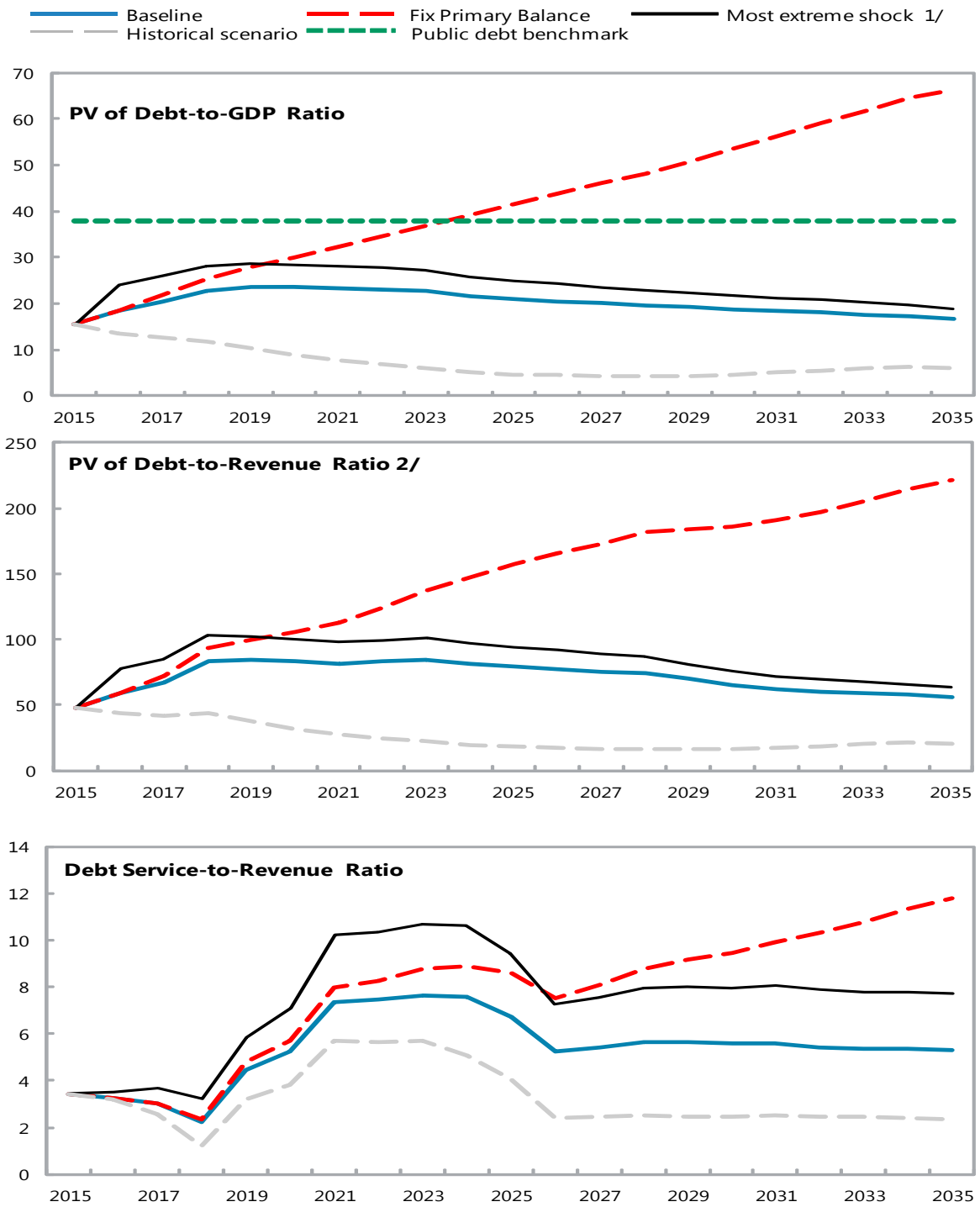
Figure 2. Liberia: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2015–35[1]



Sources: Country authorities; and staff estimates and projections.

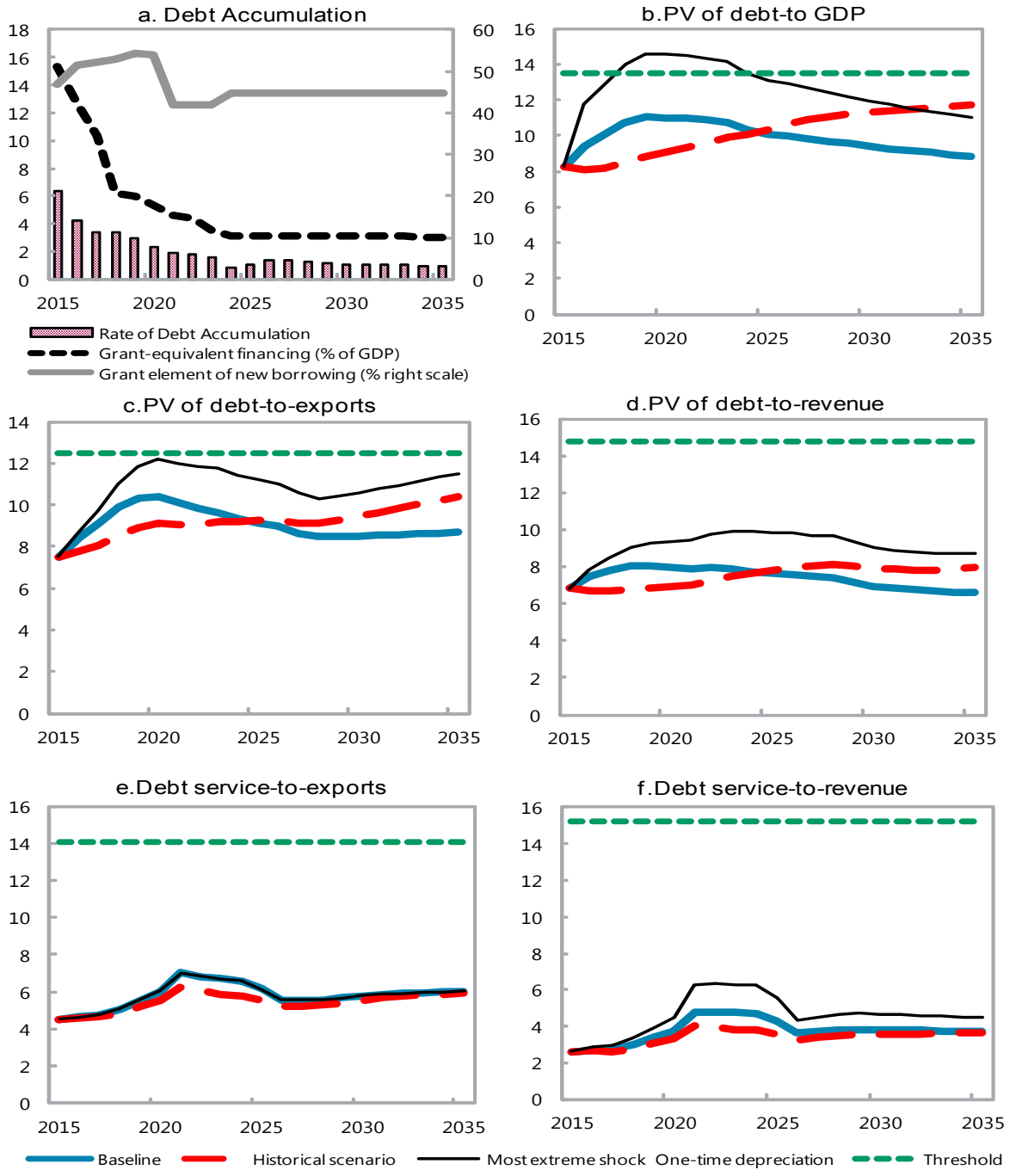
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a One-time depreciation shock; in c. to a Terms shock; in d. to a Terms shock; in e. to a Growth shock and in figure f. to a One-time depreciation shock.

Figure 3. Liberia: Indicators of Public Debt Under Alternative Scenarios, 2015–35[1]



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025.
2/ Revenues are defined inclusive of grants.

Figure 4. Liberia: Probability of Debt Distress of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2015–35[1]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a One-time depreciation shock; in c. to a Terms shock; in d. to a Terms shock; in e. to a Growth shock and in figure f. to a One-time depreciation shock.

Table 3. Liberia: External Debt Sustainability Framework, Baseline Scenario, 2012–35[1]

(Percent of GDP, fiscal year, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2012	2013	2014			2015	2016	2017	2018	2019	2020	2015-2020 Average	2025	2035	2021-2035 Average
External debt (nominal) 1/	**9.4**	**10.9**	**13.6**			**23.2**	**30.1**	**33.8**	**37.2**	**39.2**	**39.5**		**34.3**	**26.9**	
of which: public and publicly guaranteed (PPG)	9.4	10.9	13.6			23.2	30.1	33.8	37.2	39.2	39.5		34.3	26.9	
Change in external debt	-2.2	1.5	2.7			9.6	6.9	3.7	3.5	2.0	0.3		-1.4	-0.6	
Identified net debt-creating flows	-0.2	2.6	7.9			13.2	9.2	5.5	3.7	2.4	1.5		-0.5	0.3	
Non-interest current account deficit	**24.2**	**24.9**	**29.5**	**21.4**	**11.7**	**35.8**	**37.8**	**35.1**	**33.3**	**32.0**	**31.9**		**29.9**	**24.7**	29.9
Deficit in balance of goods and services	77.3	65.3	73.5			85.9	78.2	63.7	52.9	50.9	48.2		40.9	31.5	
Exports	48.3	48.4	43.0			36.3	30.4	26.8	24.5	23.5	23.1		27.2	25.0	
Imports	125.6	113.7	116.4			122.2	108.7	90.6	77.4	74.4	71.3		68.1	56.5	
Net current transfers (negative = inflow)	-66.5	-57.4	-61.6	-99.6	38.2	-65.5	-53.1	-40.4	-31.6	-29.8	-27.8		-21.1	-12.7	-18.5
of which: official	-26.7	-25.0	-32.2			-33.6	-24.7	-23.2	-22.6	-21.1	-19.5		-14.6	-8.6	
Other current account flows (negative = net inflow)	13.4	17.0	17.6			15.3	12.7	11.8	12.0	10.8	11.4		10.1	5.9	
Net FDI (negative = inflow)	**-22.8**	**-21.3**	**-20.9**	**-13.8**	**17.0**	**-23.1**	**-27.7**	**-28.6**	**-28.3**	**-27.8**	**-28.2**		**-28.6**	**-23.6**	-28.3
Endogenous debt dynamics 2/	**-1.6**	**-0.9**	**-0.8**			**0.5**	**-0.9**	**-1.1**	**-1.3**	**-1.7**	**-2.2**		**-1.8**	**-0.8**	
Contribution from nominal interest rate	0.0	0.1	0.1			0.1	0.2	0.2	0.3	0.3	0.3		0.4	0.3	
Contribution from real GDP growth	-0.8	-0.7	-0.6			0.4	-1.1	-1.3	-1.6	-2.0	-2.5		-2.2	-1.1	
Contribution from price and exchange rate changes	-0.8	-0.4	-0.3			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-2.1**	**-1.2**	**-5.2**			**-3.5**	**-2.3**	**-1.8**	**-0.2**	**-0.5**	**-1.2**		**-0.9**	**-1.0**	
of which: exceptional financing	0.0	0.0	0.0			-0.9	-0.9	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	7.9			14.2	17.9	20.0	21.9	22.8	22.7		20.1	16.2	
In percent of exports	...	...	18.5			39.3	58.8	74.4	89.1	96.7	98.4		73.8	64.8	
PV of PPG external debt	**...**	**...**	**7.9**			**14.2**	**17.9**	**20.0**	**21.9**	**22.8**	**22.7**		**20.1**	**16.2**	
In percent of exports	**...**	**...**	**18.5**			**39.3**	**58.8**	**74.4**	**89.1**	**96.7**	**98.4**		**73.8**	**64.8**	
In percent of government revenues	**...**	**...**	**33.8**			**63.7**	**79.6**	**86.5**	**90.6**	**90.6**	**89.0**		**82.6**	**58.1**	
Debt service-to-exports ratio (in percent)	**0.1**	**0.8**	**0.4**			**0.7**	**1.0**	**1.3**	**2.2**	**3.6**	**4.7**		**5.1**	**4.8**	
PPG debt service-to-exports ratio (in percent)	**0.1**	**0.8**	**0.4**			**0.6**	**1.0**	**1.3**	**2.2**	**3.6**	**4.7**		**5.1**	**4.8**	
PPG debt service-to-revenue ratio (in percent)	**0.2**	**1.4**	**0.8**			**1.1**	**1.3**	**1.6**	**2.3**	**3.3**	**4.3**		**5.7**	**4.3**	
Total gross financing need (Millions of U.S. dollars)	24.3	73.5	177.0			258.1	218.7	156.3	132.3	131.1	138.1		119.7	233.2	
Non-interest current account deficit that stabilizes debt ratio	26.5	23.4	26.8			26.2	30.9	31.5	29.8	30.0	31.6		31.3	25.3	
Key macroeconomic assumptions															
Real GDP growth (in percent)	7.9	8.5	5.7	7.1	1.9	-2.7	4.9	4.6	4.9	5.9	7.0	4.1	6.8	4.3	6.1
GDP deflator in US dollar terms (change in percent)	7.6	4.0	2.4	6.3	2.6	2.5	0.3	2.2	2.1	3.0	3.1	2.2	1.9	3.7	2.5
Effective interest rate (percent) 5/	0.1	1.6	0.7	0.4	0.6	0.9	0.8	0.8	0.8	0.9	0.9	0.8	1.2	1.2	1.2
Growth of exports of G&S (US dollar terms, in percent)	26.1	13.1	-4.0	15.1	16.8	-15.7	-11.8	-5.7	-2.2	4.6	8.4	-3.7	10.9	4.8	9.4
Growth of imports of G&S (US dollar terms, in percent)	9.3	2.2	10.7	15.9	23.9	4.7	-6.4	-10.9	-8.4	4.9	5.7	-1.8	3.7	5.0	7.1
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	46.9	51.6	52.1	52.9	54.2	53.9	51.9	44.6	44.6	44.1
Government revenues (excluding grants, in percent of GDP)	26.2	27.5	23.5			22.4	22.5	23.1	24.1	25.1	25.5		24.3	27.8	26.0
Aid flows (in Millions of US dollars) 7/	28.3	45.7	77.6			199.1	183.3	167.9	73.1	77.4	82.8		89.5	202.8	
of which: Grants	28.3	45.7	77.6			199.1	183.3	167.9	73.1	77.4	82.8		89.5	202.8	
of which: Concessional loans	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			15.4	12.6	10.4	6.2	6.0	5.4		3.1	3.0	3.3
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			71.4	77.4	79.5	68.8	70.0	71.3		69.2	70.3	68.4
Memorandum items:															
Nominal GDP (Millions of US dollars)	1643.1	1853.9	2005.5			2000.8	2103.8	2249.6	2409.9	2627.9	2898.2		4474.3	10141.8	
Nominal dollar GDP growth	16.0	12.8	8.2			-0.2	5.2	6.9	7.1	9.0	10.3	6.4	8.8	8.2	8.7
PV of PPG external debt (in Millions of US dollars)			154.3			281.1	366.5	437.1	513.4	584.2	644.4		884.8	1614.8	
(PVt-PVt-1)/GDPt-1 (in percent)						6.3	4.3	3.4	3.4	2.9	2.3	3.8	1.0	0.9	1.2
Gross workers' remittances (Millions of US dollars)	...	...	...			...	...	...	...	...	...		...	...	
PV of PPG external debt (in percent of GDP + remittances)	...	...	7.9			14.2	17.9	20.0	21.9	22.8	22.7		20.1	16.2	
PV of PPG external debt (in percent of exports + remittances)	...	...	18.5			39.3	58.8	74.4	89.1	96.7	98.4		73.8	64.8	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	0.4			0.6	1.0	1.3	2.2	3.6	4.7		5.1	4.8	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 4. Liberia: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt 2015–35

(Percent)

	Projections							
	2015	2016	2017	2018	2019	2020	**2025**	2035
PV of debt-to GDP ratio								
Baseline	14	18	20	22	23	23	**20**	16
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	14	14	14	15	16	17	**21**	25
A2. New public sector loans on less favorable terms in 2015-2035 2	14	20	23	27	29	30	**31**	29
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	14	17	19	21	22	22	**19**	16
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	14	16	18	20	21	21	**19**	15
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	14	17	19	21	22	22	**19**	15
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	14	18	14	16	17	18	**17**	15
B5. Combination of B1-B4 using one-half standard deviation shocks	14	6	-11	-9	-7	-5	**0**	8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	14	25	28	30	32	32	**28**	23
PV of debt-to-exports ratio								
Baseline	39	59	74	89	97	98	**74**	65
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	39	45	52	62	69	73	**77**	99
A2. New public sector loans on less favorable terms in 2015-2035 2	39	64	86	109	123	129	**113**	118
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	39	57	72	87	94	96	**73**	64
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	39	51	62	76	83	85	**65**	59
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	39	57	72	87	94	96	**73**	64
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	39	60	53	67	74	77	**61**	59
B5. Combination of B1-B4 using one-half standard deviation shocks	39	18	-38	-32	-25	-20	**1**	30
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	39	57	72	87	94	96	**73**	64
PV of debt-to-revenue ratio								
Baseline	64	80	86	91	91	89	**83**	58
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	64	61	60	63	65	66	**86**	89
A2. New public sector loans on less favorable terms in 2015-2035 2	64	87	100	110	115	117	**126**	106
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	64	77	83	87	87	86	**80**	56
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	64	73	76	81	82	81	**77**	56
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	64	75	81	85	86	84	**79**	55
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	64	81	62	68	70	70	**68**	53
B5. Combination of B1-B4 using one-half standard deviation shocks	64	27	-49	-36	-26	-20	**1**	30
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	64	110	120	126	126	124	**116**	81

Table 4. Liberia: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt 2015–35 (Concluded)

(Percent)

	Projections							
	2015	2016	2017	2018	2019	2020	**2025**	2035
Debt service-to-exports ratio								
Baseline	1	1	1	2	4	5	**5**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	1	1	1	2	3	4	**4**	5
A2. New public sector loans on less favorable terms in 2015-2035 2	1	1	1	2	4	6	**5**	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	1	1	1	2	4	5	**5**	5
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	1	1	1	2	3	4	**5**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	1	1	1	2	4	5	**5**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	1	1	1	2	3	4	**4**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	1	1	0	-1	1	2	**0**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	1	1	1	2	4	5	**5**	5
Debt service-to-revenue ratio								
Baseline	1	1	2	2	3	4	**6**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	1	1	1	2	2	3	**4**	4
A2. New public sector loans on less favorable terms in 2015-2035 2	1	1	2	2	4	5	**6**	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	1	1	2	2	3	4	**6**	4
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	1	1	1	2	3	4	**5**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	1	1	2	2	3	4	**6**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	1	1	2	2	3	4	**5**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	1	1	0	-1	1	2	**0**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	1	2	2	3	5	6	**8**	6
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	42	42	42	42	42	42	**42**	42

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 5. Liberia: Public Sector Debt Sustainability Framework, Baseline Scenario, 2015–35

(Percent of GDP, fiscal year, unless otherwise indicated)

	Actual					Estimate	Projections								
	2012	2013	2014	Average 5/	Standard Deviation 5/	2015	2016	2017	2018	2019	2020	2015-20 Average	2025	2035	2021-35 Average
Public sector debt 1/	11.0	12.4	16.1			24.4	30.7	34.2	38.1	40.0	40.3		35.1	27.3	
of which: foreign-currency denominated	11.0	12.4	16.1			24.4	30.7	34.2	38.1	40.0	40.3		35.1	27.3	
Change in public sector debt	-2.6	1.4	3.8			8.3	6.3	3.5	3.9	2.0	0.3		-1.4	-1.0	
Identified debt-creating flows	1.3	0.6	1.0			7.9	7.6	3.2	3.6	1.8	0.7		-1.7	-2.3	
Primary deficit	3.2	1.4	1.8	0.0	2.1	7.9	8.2	4.9	5.7	4.7	4.1	5.9	0.9	-0.5	0.8
Revenue and grants	27.9	29.9	27.4			32.3	31.2	30.6	27.2	28.1	28.4		26.3	29.8	
of which: grants	1.7	2.5	3.9			10.0	8.7	7.5	3.0	2.9	2.9		2.0	2.0	
Primary (noninterest) expenditure	31.2	31.4	29.2			40.2	39.5	35.4	32.8	32.8	32.5		27.2	29.4	
Automatic debt dynamics	-1.9	-0.8	-0.8			0.0	-0.6	-1.7	-2.0	-3.0	-3.5		-2.6	-1.8	
Contribution from interest rate/growth differential	-1.2	-0.9	-0.7			0.5	-1.2	-1.6	-1.9	-2.6	-3.1		-2.6	-1.4	
of which: contribution from average real interest rate	-0.2	0.0	0.0			0.0	0.0	-0.2	-0.3	-0.5	-0.5		-0.3	-0.2	
of which: contribution from real GDP growth	-1.0	-0.9	-0.7			0.4	-1.1	-1.4	-1.6	-2.1	-2.6		-2.3	-1.2	
Contribution from real exchange rate depreciation	-0.7	0.1	-0.1			-0.5	0.6	-0.1	-0.1	-0.4	-0.4		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-3.9	0.8	2.8			0.4	-1.3	0.3	0.2	0.2	-0.4		0.3	1.3	
Other Sustainability Indicators															
PV of public sector debt	...	...	10.5			15.5	18.5	20.4	22.7	23.6	23.5		20.9	16.5	
of which: foreign-currency denominated	...	...	10.5			15.5	18.5	20.4	22.7	23.6	23.5		20.9	16.5	
of which: external	...	...	7.9			14.2	17.9	20.0	21.9	22.8	22.7		20.1	16.2	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	3.4	2.3	3.0			9.0	9.2	5.8	6.3	6.0	5.6		2.7	1.1	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	38.3			47.9	59.3	66.9	83.5	84.0	82.9		79.3	55.5	
PV of public sector debt-to-revenue ratio (in percent)	...	...	44.5			69.2	82.3	88.5	94.0	93.8	92.2		85.9	59.5	
of which: external 3/	...	...	33.8			63.7	79.6	86.5	90.6	90.6	89.0		82.6	58.1	
Debt service-to-revenue and grants ratio (in percent) 4/	0.4	3.0	4.2			3.5	3.2	3.0	2.2	4.4	5.3		6.8	5.3	
Debt service-to-revenue ratio (in percent) 4/	0.5	3.2	4.9			5.0	4.5	4.0	2.5	5.0	5.9		7.3	5.7	
Primary deficit that stabilizes the debt-to-GDP ratio	5.8	0.1	-2.0			-0.4	1.9	1.4	1.8	2.8	3.9		2.3	0.6	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	7.9	8.5	5.7	7.1	1.9	-2.7	4.9	4.6	4.9	5.9	7.0	4.1	6.8	4.3	6.1
Average nominal interest rate on forex debt (in percent)	0.2	1.4	1.3	0.6	0.6	1.4	1.1	0.9	1.0	0.8	0.8	1.0	1.2	1.2	1.1
Average real interest rate on domestic debt (in percent)	-5.7	-6.6	...	-5.9	4.4	...	...	...	...	...	...	...	...	...	...
Real exchange rate depreciation (in percent, + indicates depreciation)	-5.6	0.6	-1.1	-3.6	4.0	-2.9	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	8.7	8.0	9.2	9.5	3.5	7.3	4.7	8.2	7.8	8.2	8.0	7.3	5.2	6.9	6.0
Growth of real primary spending (deflated by GDP deflator, in percent)	24.5	9.1	-1.7	3.4	8.0	34.2	2.9	-6.0	-2.9	5.8	6.1	6.7	5.5	3.8	5.4
Grant element of new external borrowing (in percent)	...	...	...	...	...	46.9	51.6	52.1	52.9	54.2	53.9	51.9	44.6	44.6	...

Sources: Country authorities; and staff estimates and projections.

1/ Central government gross debt.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 6. Liberia: Sensitivity Analysis for Key Indicators of Public Debt, 2015–35

(Percent)

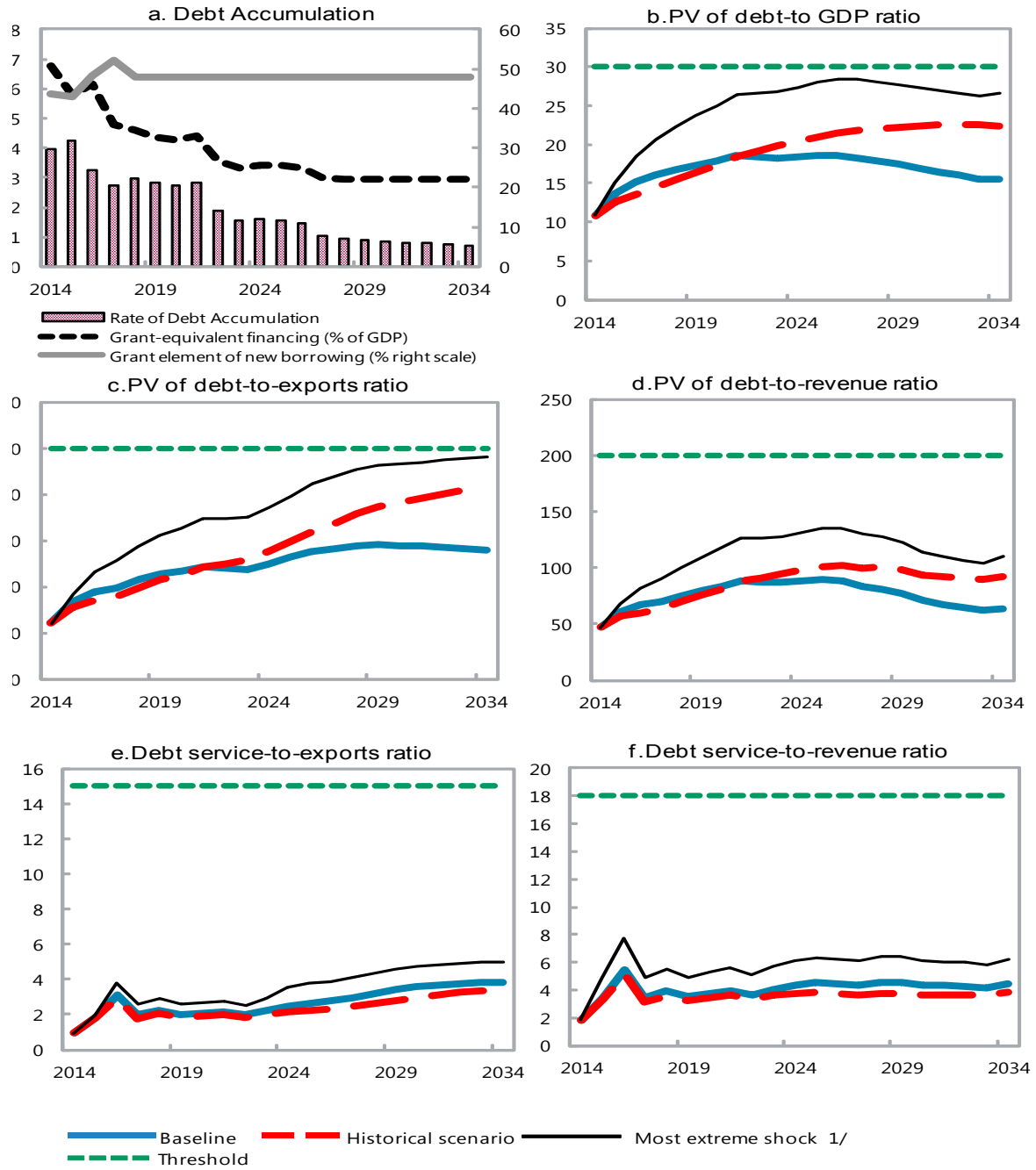
	Projections							
	2015	2016	2017	2018	2019	2020	2025	2035
PV of Debt-to-GDP Ratio								
Baseline	15	19	20	23	24	24	21	17
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	15	14	13	12	10	9	5	6
A2. Primary balance is unchanged from 2015	15	18	22	25	28	30	41	66
A3. Permanently lower GDP growth 1/	15	19	21	23	25	25	25	28
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	15	18	20	22	23	23	20	15
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	15	15	15	18	19	19	18	15
B3. Combination of B1-B2 using one half standard deviation shocks	15	14	14	16	17	17	14	9
B4. One-time 30 percent real depreciation in 2016	15	24	25	26	26	25	22	18
B5. 10 percent of GDP increase in other debt-creating flows in 2016	15	24	26	28	29	28	25	19
PV of Debt-to-Revenue Ratio 2/								
Baseline	48	59	67	84	84	83	79	55
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	48	44	42	43	37	31	18	21
A2. Primary balance is unchanged from 2015	48	59	72	94	100	106	158	222
A3. Permanently lower GDP growth 1/	48	60	68	86	87	87	93	93
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	48	59	66	82	82	81	75	50
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	48	48	51	66	68	68	68	51
B3. Combination of B1-B2 using one half standard deviation shocks	48	46	46	59	60	59	53	31
B4. One-time 30 percent real depreciation in 2016	48	77	81	95	91	88	82	59
B5. 10 percent of GDP increase in other debt-creating flows in 2016	48	77	85	103	102	100	94	63
Debt Service-to-Revenue Ratio 2/								
Baseline	3	3	3	2	4	5	7	5
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	3	3	3	1	3	4	4	2
A2. Primary balance is unchanged from 2015	3	3	3	2	5	6	9	12
A3. Permanently lower GDP growth 1/	3	3	3	2	5	5	7	7
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	3	3	3	2	4	5	7	5
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	3	3	3	2	4	5	6	5
B3. Combination of B1-B2 using one half standard deviation shocks	3	3	3	1	4	5	5	4
B4. One-time 30 percent real depreciation in 2016	3	4	4	3	6	7	9	8
B5. 10 percent of GDP increase in other debt-creating flows in 2016	3	3	4	3	5	6	8	6

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.

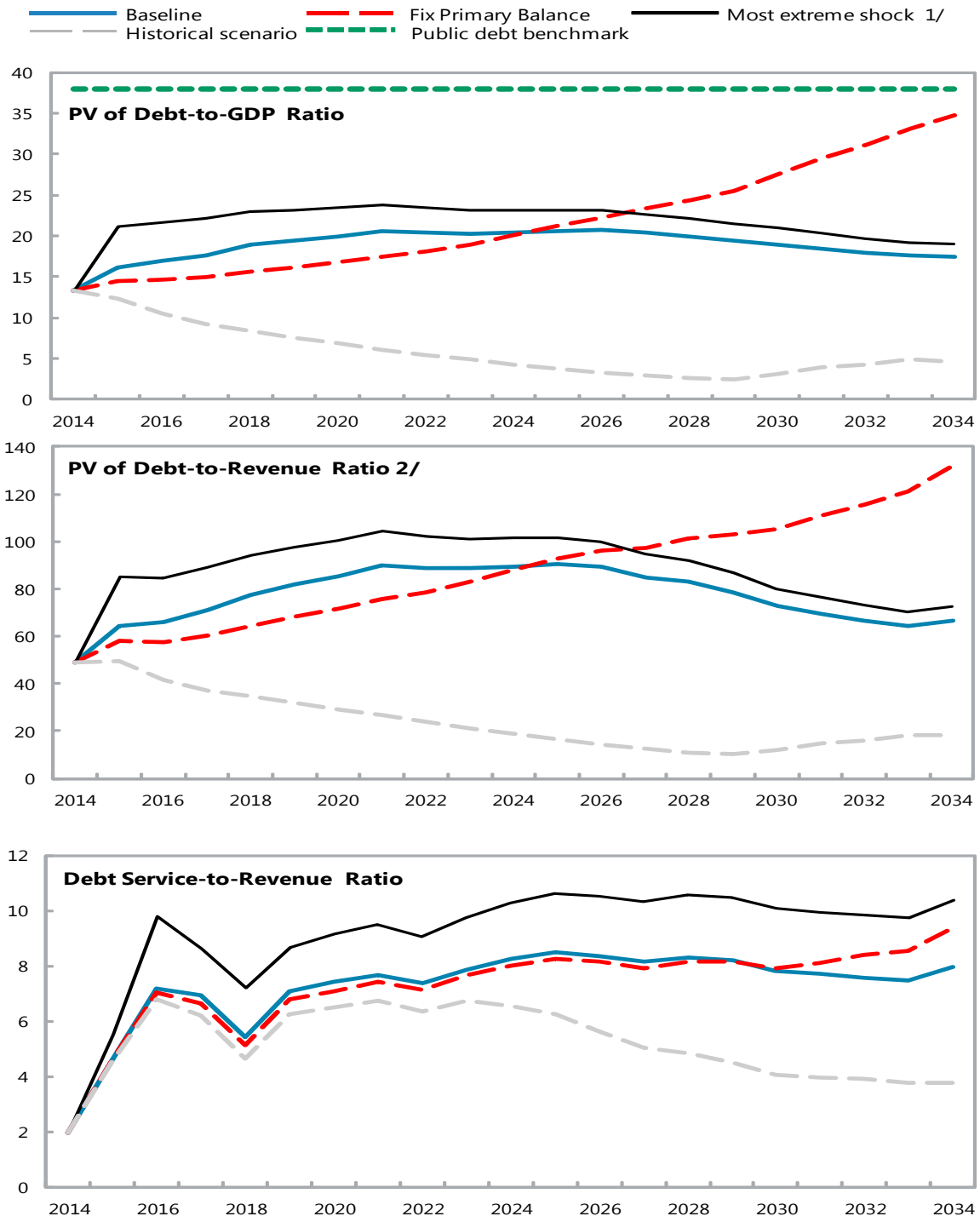
Figure 5. Liberia: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios (DSA for the 3rd ECF Review in June 2014), 2014–34[1]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Terms shock; in c. to a Terms shock; in d. to a Terms shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock. In the alternative scenarios of external DSA, historical average and standard deviation of major variables are calculated by using the data from 2008 due to the structual change of the economy.

Figure 6. Liberia: Indicators of Public Debt Under Alternatives Scenarios (DSA for the 3rd ECF Review in June 2014), 2014–34[1/]



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.